Poststreptococcal Glomerulonephritis

- ☑ Affects children 5–12 years of age; rare below 3 years.
- ☑ Presentation after 1–2 weeks after sore throat or 2–4 weeks after impetigo; nephritogenic strains 4, 12, 25, and 49 are most often implicated. Clinical presentation may be variable **(Table 2)**.
- ☑ Pathological hallmarks include diffuse proliferation of glomerular endothelial and mesangial cells, neutrophilic infiltration, granular capillary wall and mesangial deposits of immunoglobulin G (IgG) and C3 on immunofluorescence and subepithelial deposits (humps) on electron microscopy.

TABLE 2: Clinical presentation of PSGN.

Acute nephritic syndrome (most common)	Active urinary sediment (red cell casts, white blood cells, and dysmorphic red blood cells); proteinuria (usually <1 g/m^2/day); gross hematuria/cola colored urine in 30–70%; variable degrees of hypertension in 60–80%; and acute kidney injury
Subclinical disease (during epidemics)	Asymptomatic microscopic hematuria (>5 red blood cells per high power field of centrifuged urine specimen), or proteinuria (<40 mg/m^2/day); blood pressure may be normal or transiently increased; normal kidney function
Atypical presentation	Acute pulmonary edema, hypertensive encephalopathy, nephrotic syndrome (proteinuria >1 g/m^2/day, serum albumin <3 g/dL, edema), anuric AKI, rapidly progressive renal failure (acute GN with rapid decline in kidney function within 7–10 days)

(AKI: acute kidney injury; GN: poststreptococcal glomerulonephritis; PSGN: poststreptococcal glomerulonephritis)

Investigations

- ☑ *Urinalysis:* Urine microscopy (cells and red blood cell casts)
- ☑ Complete hemogram [complete blood count (CBC)], erythrocyte sedimentation rate (ESR), and C-reactive protein (CRP)
- ☑ Early morning spot urine protein to creatinine ratio (UPCR)/24-hour urine protein and creatinine
- ☑ *Biochemistry:* Blood urea, serum creatinine, serum electrolytes, albumin, and cholesterol
- ☑ Antistreptolysin O tier [elevated in 60–80% of poststreptococcal glomerulonephritis (PSGN)] and antideoxyribonuclease B
- ☑ Complement C3 (90% low C3 in PSGN, normalizes in 6–8 weeks)
- ☑ Antinuclear antibodies (ANAs), antineutrophil cytoplasmic antibodies (c-ANCA, p-ANCA) as indicated
- ☑ Hepatitis B, C and HIV serology

Investigations

Kidney Biopsy

Kidney biopsy is not required for typical PSGN. It is done to assess prognosis in severe forms with anuric acute kidney injury (AKI)/rapidly progressive renal failure (RPRF) or diagnose other etiology of acute GN **(Box 1)**.

BOX 1: Indications for kidney biopsy in acute glomerulonephritis (GN).

At onset:

- ☑ *Systemic features:* Fever, rash, joint pain, and heart disease
- ☑ Normal C3 levels
- ☑ Atypical presentation (see above)

Delayed resolution:

- ☑ Persisting oliguria, hypertension, or reduced kidney function >10–14 days
- ☑ Gross hematuria >3–4 weeks
- ☑ Low C3 complement >12 weeks
- ☑ Persisting subnephrotic range proteinuria >6 months
- ☑ Persisting microscopic hematuria >12–18 months

Management of Poststreptococcal Glomerulonephritis

- ☑ Symptomatic management; hospitalization for severe hypertension and oliguria
- ☑ Monitoring daily weight, fluid balance, blood levels of urea, creatinine, and electrolytes
- ☑ Fluid and salt restriction (<1.0 g/day) and judicious use of diuretics (furosemide 1–2 mg/kg/day) if oliguria and edema
- ☑ *Hypertension:*
  - Moderate hypertension treated with diuretics and antihypertensive agents (calcium channel blockers)
  - In hypertensive emergency, drug of choice is intravenous (IV) labetalol (0.5–1 mg/kg/h); sodium nitroprusside (0.3–8 µg/kg/minute)
  - Short acting nifedipine (0.25–0.5 mg/kg/dose) should be used with caution
  - Avoid angiotensin converting enzyme inhibitors due to risk of hyperkalemia
- ☑ *Hyperkalemia*:
  - No added potassium in diet and fluids
  - Medical management of hyperkalemia
  - Dialysis in refractory hyperkalemia

Management of Poststreptococcal Glomerulonephritis

- ☑ *Acute kidney injury*: Correct fluid and electrolyte imbalance, dialysis if indicated.
  - *Indications of dialysis*: Fluid overload, pulmonary edema, uncontrolled hypertension, severe hyperkalemia, uremia with seizures/encephalopathy, and refractory metabolic acidosis
- ☑ *Rapidly progressive GN (RPGN)*: Dialysis as per the indication and early kidney biopsy for etiological diagnosis and definitive management. The histologic hallmark of RPGN is crescentic GN. Initiation of immunosuppression with IV pulse methylprednisolone followed by oral prednisolone improves the outcome. IV cyclophosphamide (five to six doses) may additionally be considered in systemic vasculitis. Specific management of infection-related GN (IRGN) other than PSGN is given in **Table 3**.

TABLE 3: Specific management of IRGN other than PSGN.

Infected shunts and endocarditis	Appropriate antibiotics and removal of shunts/prosthetic devices
Hepatitis B or C associated GN	Antiviral drugs as indicated
Other virus: HIV	Antiretroviral therapy and ACE-inhibition

(ACE: angiotensin-converting enzyme; GN: glomerulonephritis; HIV: human immunodeficiency virus; IRGN: infection-related GN; PSGN: poststreptococcal GN)

Red Flag Signs in Infection-related Glomerulonephritis

- ☑ Age < 3 years
- ☑ Anuric AKI
- ☑ Rapidly progressive kidney failure
- ☑ AKI requiring dialysis
- ☑ Nephritic nephrotic syndrome
- ☑ Crescents on kidney biopsy
- ☑ Persistent abnormalities on follow-up (high blood pressure, edema, proteinuria, active urinary sediments, and persistently low C3)

Prognosis

- ☑ Prognosis of PSGN in acute phase and in long term is excellent.
- ☑ Mortality is <1%.
- ☑ Urinary abnormalities present in about 20% children on long term.
- ☑ Infection-related GN other than postinfectious GN is variable and guarded; hypertension, proteinuria, and progressive kidney disease are often found on long-term follow-up.

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